

8. Generative Adversarial Network

Young-geun Kim

Department of Mathematical Sciences

MTH345: Deep Generative Models: A Statistical Perspective



ULSAN NATIONAL INSTITUTE OF
SCIENCE AND TECHNOLOGY

Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

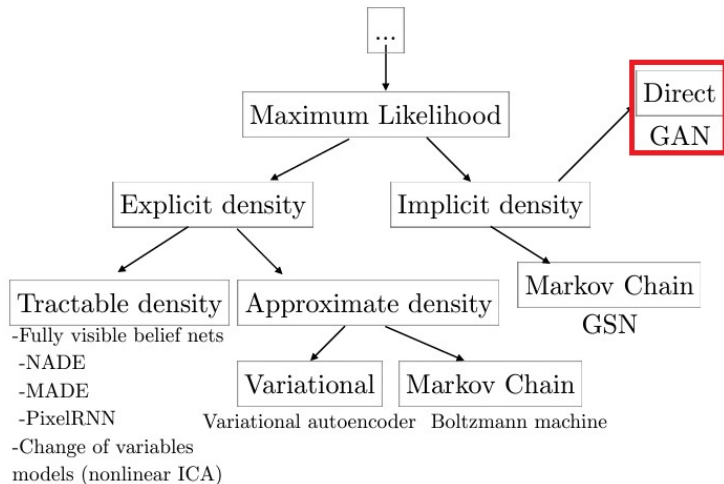
M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics



The figure is edited from Goodfellow (2016).

Lecture Topics

- ▶ Recommended reference materials include:
 - ▶ Goodfellow et al. (2014), “Generative adversarial nets,” *Advances in neural information processing systems*.
 - ▶ Goodfellow (2016), “Nips 2016 tutorial: Generative adversarial networks,” *arXiv*.
- ▶ It is also encouraged to refer to *Chapter 8. Generative Adversarial Networks* in Kim (2026). Compared with that reference, this lecture note includes additional background on adversarial learning and covers roughly 30% of the GAN research addressed in the cited material.

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2. Representation Learning with Adversarial Learning
3. Data-to-Data Translation
4. Extension to f -divergence

Motivation

- ▶ Most previously reviewed methods, such as ICA, auto-regressive models, RBMs, and VAEs, explicitly formulate density models.
- ▶ However, due to non-linear generators, the likelihoods are often less tractable, which necessitates the use of surrogate density forms.
- ▶ RBMs utilize the contrastive divergence algorithm:

$$\frac{\partial \log p_{\theta}(\mathbf{x})}{\partial \theta} = - \sum_{\mathbf{z}} \left(\frac{\partial E_{\theta}(\mathbf{z}, \mathbf{x})}{\partial \theta} \right) p_{\theta}(\mathbf{z}|\mathbf{x}) + \sum_{\mathbf{z}, \mathbf{x}} \left(\frac{\partial E_{\theta}(\mathbf{z}, \mathbf{x})}{\partial \theta} \right) p_{\theta}(\mathbf{z}, \mathbf{x}) \quad (1)$$

where each term is approximated using a Markov chain.

- ▶ VAEs utilizes the variational inference:

$$\log p_{\theta}(\mathbf{x}) - \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p_{\theta}(\mathbf{z}|\mathbf{x})). \quad (2)$$

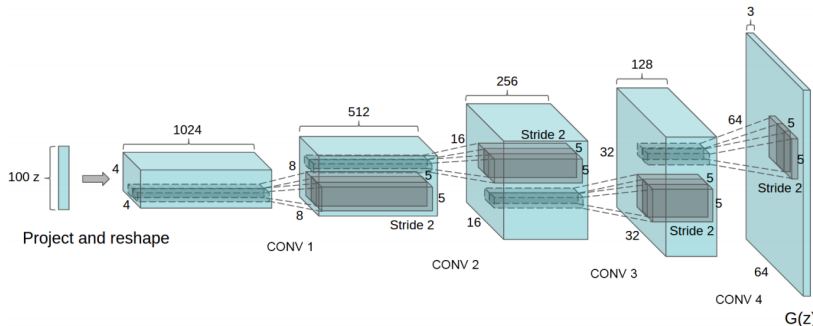
Motivation



- ▶ The approximations of (gradients of) log-likelihoods often fail, resulting in blurry generation results.
- ▶ Generative Adversarial Networks (GANs; Goodfellow et al., 2014) propose an alternative approach to approximating a likelihood ratio-based objective.

Images are from Tolstikhin et al. (2018).

Generative Adversarial Network



- ▶ GANs introduce adversarial learning through two networks: (i) a generator and (ii) a discriminator.
- ▶ The generator specifies the same graphical model used in VAEs, but $\mathbf{x} = G_{\theta}(\mathbf{z})$ where G is a neural network, i.e., $p_{\theta}(\mathbf{x}|\mathbf{z})$ degenerates to a single point.

Example of the generator architecture of deep convolutional GANs (Radford et al., 2016).

Adversarial Learning

- ▶ In adversarial learning, the key idea is to train a binary classifier that distinguishes real data from generated ones, and to update the generator ($G_\theta(\mathbf{z})$) to fool it.
- ▶ The discriminator is trained to classify data from the following two classes:

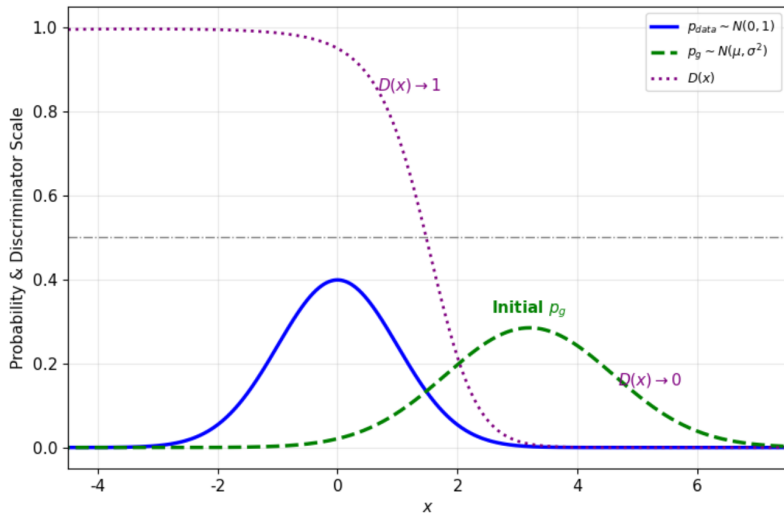
$$(\text{Real Data}; Y = 1) \mathbf{x}_1, \dots, \mathbf{x}_n \sim p(\mathbf{x}) \quad (3)$$

and

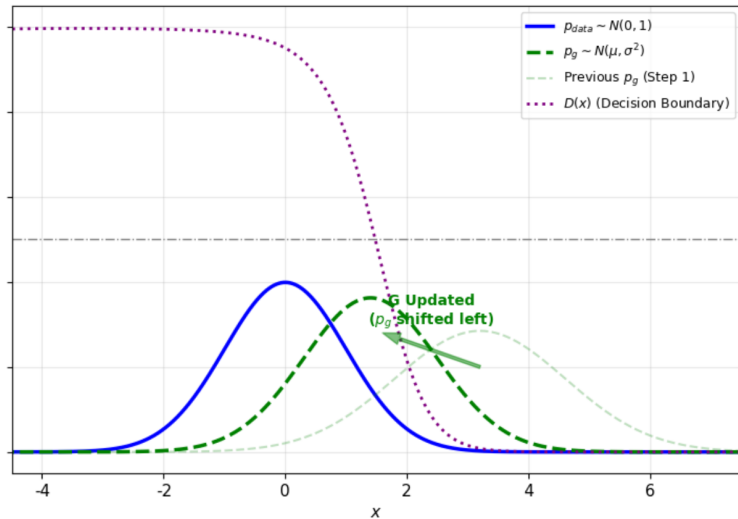
$$(\text{Generated Data}; Y = 0) G_\theta(\mathbf{z}_1), \dots, G_\theta(\mathbf{z}_n), \text{ where } \mathbf{z}_1, \dots, \mathbf{z}_n \sim p(\mathbf{z}). \quad (4)$$

- ▶ We denote the output of the classifier (or *discriminator*) as $D_\phi(\mathbf{x})$, which models $P(Y = 1 | \mathbf{X} = \mathbf{x})$.
- ▶ The generator G_θ is then updated to maximize the loss of the discriminator D_ϕ . This process is iteratively repeated, which is referred to as *adversarial learning*.

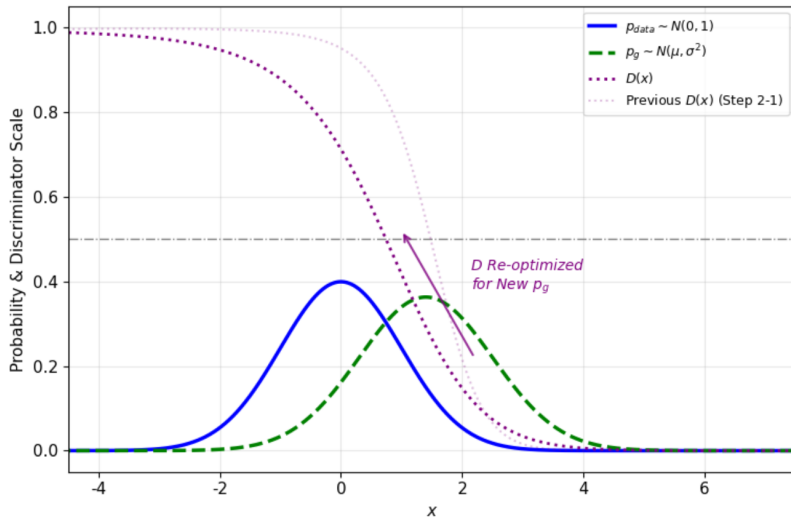
Adversarial Learning



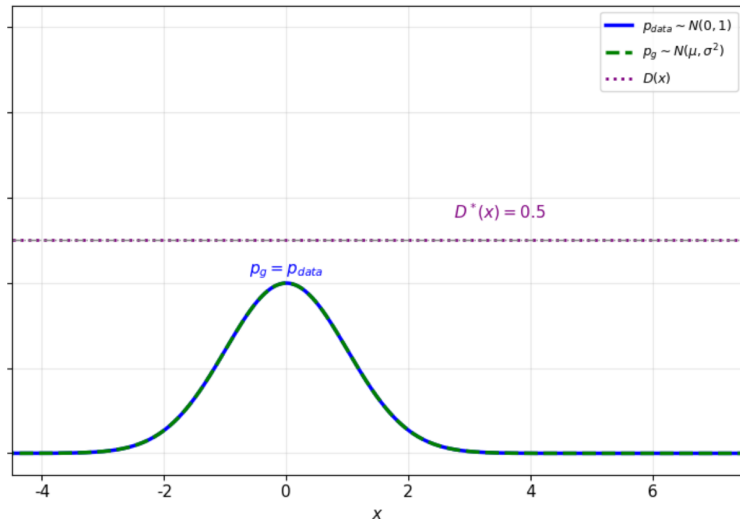
Adversarial Learning



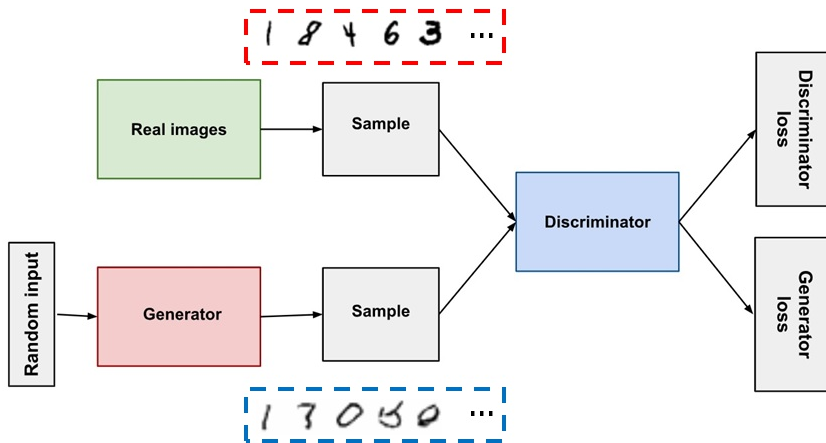
Adversarial Learning



Adversarial Learning



Adversarial Learning



Images were edited from
https://developers.google.com/machine-learning/gan/gan_structure and
<https://github.com/MorvanZhou/mnistGANs>.

Adversarial Learning

- ▶ Adversarial learning can be formulated as follows.
- ▶ First, the loss function of D_ϕ is the negative log-likelihood (or *cross-entropy loss*):

$$l(D_\phi(\mathbf{x}), y) = -l(y = 1) \log D_\phi(\mathbf{x}) - l(y = 0) \log (1 - D_\phi(\mathbf{x})). \quad (5)$$

- ▶ With the dataset consisting of n real data and n generated data, $(\mathbf{x}_1, y_1 = 1), \dots, (\mathbf{x}_n, y_n = 1), (G_\theta(\mathbf{z}_1), y_{n+1} = 0), \dots, (G_\theta(\mathbf{z}_n), y_{2n} = 0)$, the average loss (or *empirical risk*) can be expressed as:

$$-n^{-1} \sum_{i=1}^n \log D_\phi(\mathbf{x}_i) - n^{-1} \sum_{i=1}^n \log(1 - D_\phi(G_\theta(\mathbf{z}_i))). \quad (6)$$

- ▶ Next, the loss function of the generator G_θ is $-l(D_\phi(G_\theta(\mathbf{z})), y = 0)$, and the objective function can be expressed as:

$$n^{-1} \sum_{i=1}^n \log(1 - D_\phi(G_\theta(\mathbf{z}_i))). \quad (7)$$

Adversarial Learning

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k , is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**

for k steps **do**

- Sample minibatch of m noise samples $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(m)}\}$ from noise prior $p_g(\mathbf{z})$.
- Sample minibatch of m examples $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}\}$ from data generating distribution $p_{\text{data}}(\mathbf{x})$.
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D(\mathbf{x}^{(i)}) + \log \left(1 - D(G(\mathbf{z}^{(i)})) \right) \right].$$

end for

- Sample minibatch of m noise samples $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(m)}\}$ from noise prior $p_g(\mathbf{z})$.
- Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log \left(1 - D(G(\mathbf{z}^{(i)})) \right).$$

end for

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

The pseudo-algorithm is from Goodfellow et al. (2014).

Adversarial Learning

- ▶ The adversarial learning process involves alternately maximizing and minimizing the negative cross-entropy loss:

$$V(\theta, \phi) := \int (\log D_{\phi}(\mathbf{x})) p(\mathbf{x}) d\mathbf{x} + \int (\log(1 - D_{\phi}(\mathbf{x}))) p_{\theta}(\mathbf{x}) d\mathbf{x}. \quad (8)$$

- ▶ This process can be viewed as a two-player minimax game where the goal is to find

$$\theta^* \in \arg \min_{\theta} (\max_{\phi} V(\theta, \phi)). \quad (9)$$

- ▶ The adversarial learning can be interpreted as repeated cycles of approximating and minimizing the Jensen-Shannon (JS) divergence:

$$\max_{\phi} V(\theta, \phi) = \mathcal{D}_{\text{JS}}(p \parallel p_{\theta}) := \frac{1}{2} \left(\mathcal{D}_{\text{KL}} \left(p \parallel \frac{p + p_{\theta}}{2} \right) + \mathcal{D}_{\text{KL}} \left(p_{\theta} \parallel \frac{p + p_{\theta}}{2} \right) \right). \quad (10)$$

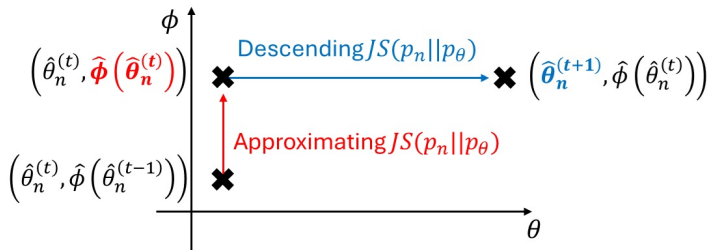
up to a constant addition and sign-preserving multiplication.

Adversarial Learning

- ▶ We assume that the discriminator class $\{D_\phi \mid \phi \in \Phi\}$ is sufficiently flexible such that for any given θ , there exists a $\phi^*(\theta)$ where:

$$D_{\phi^*(\theta)}(\mathbf{x}) = \frac{p(\mathbf{x})}{p(\mathbf{x}) + p_\theta(\mathbf{x})}. \quad (11)$$

Adversarial Learning



- The two adversarial networks, the discriminator and the generator, are trained alternately:
 1. Given $\hat{\theta}_n^{(t)}$, update the discriminator to obtain $\hat{\phi}(\hat{\theta}_n^{(t)})$ by maximizing $V(\hat{\theta}_n^{(t)}, \phi)$.
 2. Given $\hat{\phi}(\hat{\theta}_n^{(t)})$, update the generator to obtain $\hat{\theta}_n^{(t+1)}$ by minimizing $V(\theta, \hat{\phi}(\hat{\theta}_n^{(t)}))$.
 3. Repeat the above processes.

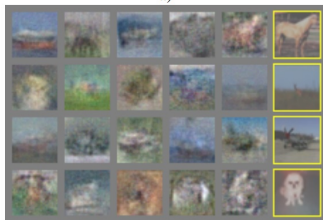
Generation Result



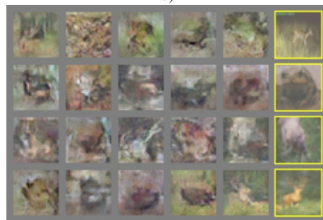
a)



b)



c)



d)

Images are from Goodfellow et al. (2014).

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Representation Learning with Adversarial Learning

- ▶ In (vanilla) GANs, there is no density-based inference, as they do not have explicit density models.
- ▶ Several variants of GANs have been proposed to learn (interpretable) representations and conduct corresponding inferences.

Bidirectional Generative Adversarial Network

- ▶ Bidirectional GANs (BiGANs) (Donahue et al., 2016) introduce an encoder $E_\psi(\mathbf{x})$ to directly learn the dimension-reduction mapping.¹
- ▶ The discriminator is modified to classify pairs $(\mathbf{X}, E_\psi(\mathbf{X}))$ vs. $(G_\theta(\mathbf{Z}), \mathbf{Z})$. The corresponding value function can be expressed as:

$$V(\theta, \phi, \psi) := \int (\log D_\phi(\mathbf{x}, E_\psi(\mathbf{x}))) p(\mathbf{x}) d\mathbf{x} + \int (\log(1 - D_\phi(G_\theta(\mathbf{z}), \mathbf{z}))) p(\mathbf{z}) d\mathbf{z}. \quad (12)$$

- ▶ The maximized value function over ϕ is $\mathcal{D}_{JS}(p(\mathbf{x}, E_\psi(\mathbf{x})) || p_\theta(G_\theta(\mathbf{z}), \mathbf{z}))$, and is minimized w.r.t. encoder and generator parameters (θ, ψ) .
- ▶ The optimal encoder and generator identify an ideal autoencoder that perfectly reconstruct $\mathbf{x}$.

¹Adversarially Learned Inference (Dumoulin et al., 2016) is a concurrently proposed work with a similar idea.

Bidirectional Generative Adversarial Network

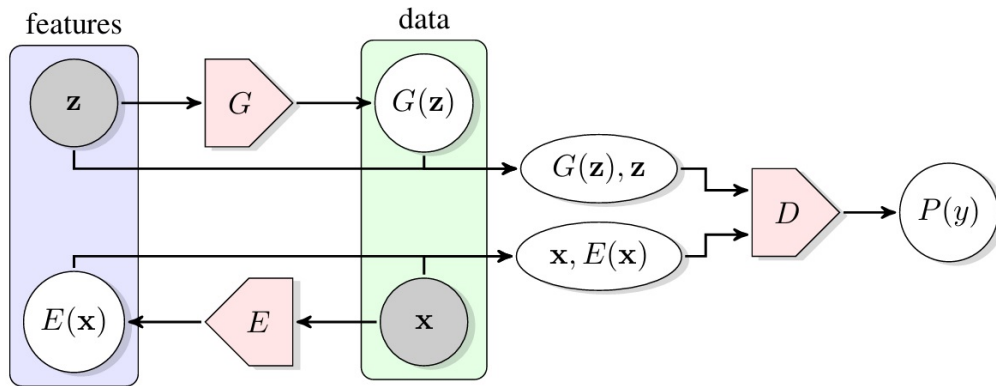
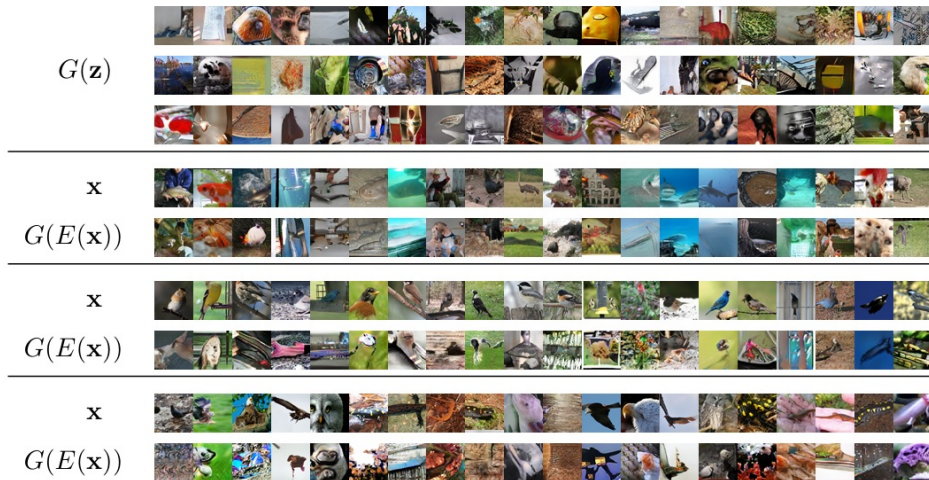


Figure 1: The structure of Bidirectional Generative Adversarial Networks (BiGAN).

Images are from Donahue et al. (2016).

Bidirectional Generative Adversarial Network



Images are from Donahue et al. (2016).

Adversarially Learned Anomaly Detection

- ▶ Despite the optimal encoder and generator perfectly reconstruct $\mathbf{x}$, it has been observed that this may empirically not hold (Li et al., 2017).
- ▶ Adversarially Learned Anomaly Detection (Zenati et al., 2018) introduces adversarial learning about the reconstructed results:

$$\begin{aligned}
 V(\theta, \phi, \psi) &:= \int (\log D_{XZ, \phi}(\mathbf{x}, E_{\psi}(\mathbf{x}))) p(\mathbf{x}) d\mathbf{x} + \int (\log(1 - D_{XZ, \phi}(G_{\theta}(\mathbf{z}), \mathbf{z}))) p(\mathbf{z}) d\mathbf{z} \\
 &+ \int (\log D_{XX, \phi}(\mathbf{x}, \mathbf{x})) p(\mathbf{x}) d\mathbf{x} + \int (\log(1 - D_{XX, \phi}(\mathbf{x}, G_{\theta}(E_{\psi}(\mathbf{x})))) p(\mathbf{x}) d\mathbf{x} \\
 &+ \int (\log D_{ZZ, \phi}(\mathbf{z}, \mathbf{z})) p(\mathbf{z}) d\mathbf{z} + \int (\log(1 - D_{ZZ, \phi}(\mathbf{z}, E_{\psi}(G_{\theta}(\mathbf{z})))) p(\mathbf{z}) d\mathbf{z}.
 \end{aligned} \tag{13}$$

Here, $D_{XZ, \phi}$, $D_{XX, \phi}$, and $D_{ZZ, \phi}$ are all discriminators.

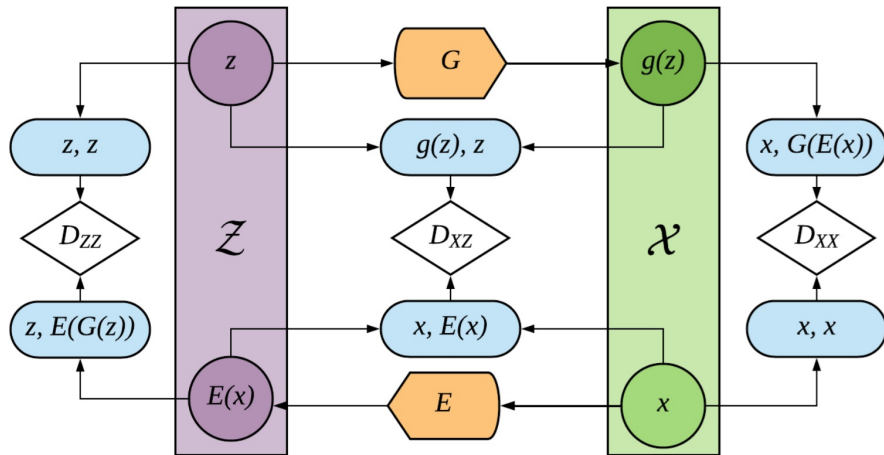
Adversarially Learned Anomaly Detection

- ▶ The maximized value function over the discriminator parameter ϕ is:

$$\begin{aligned}\max_{\phi} V(\theta, \phi, \psi) = & (2\mathcal{D}_{\text{JS}}(p(\mathbf{x}, E_{\psi}(\mathbf{x})) || p_{\theta}(G_{\theta}(\mathbf{z}), \mathbf{z})) - \log 4) \\ & + (2\mathcal{D}_{\text{JS}}(p(\mathbf{x}, \mathbf{x}) || p_{\theta, \psi}(\mathbf{x}, G_{\theta}(E_{\psi}(\mathbf{x})))) - \log 4) \\ & + (2\mathcal{D}_{\text{JS}}(p(\mathbf{z}, \mathbf{z}) || p_{\theta, \psi}(\mathbf{z}, E_{\psi}(G_{\theta}(\mathbf{z})))) - \log 4).\end{aligned}\tag{14}$$

- ▶ The optimal encoder and generator still identify an ideal autoencoder.

Adversarially Learned Anomaly Detection



Images are from Zenati et al. (2018).

Adversarially Learned Anomaly Detection

Algorithm 1 Adversarially Learned Anomaly Detection

Input $\mathbf{x}, \sim p_{\mathcal{X}_{Test}}(\mathbf{x}), E, G, f_{xx}$ where f_{xx} is the feature layer of D_{xx}

Output $A(\mathbf{x})$, where A is the anomaly score

1: **procedure** INFERENCE

2: $\tilde{\mathbf{z}} \leftarrow E(\mathbf{x})$

▷ Encode samples

3: $\hat{\mathbf{x}} \leftarrow G(\tilde{\mathbf{z}}),$

▷ Reconstruct samples

4: $f_{\delta} \leftarrow f_{xx}(\mathbf{x}, \hat{\mathbf{x}})$

5: $f_{\alpha} \leftarrow f_{xx}(\mathbf{x}, \mathbf{x})$

6: return $\|f_{\delta} - f_{\alpha}\|_1$

7: **end procedure**

The algorithm is from Zenati et al. (2018).

Adversarially Learned Anomaly Detection



Figure 5. Error analysis on SVHN when the normal class is the “3” digit. 1st row: normal data; 2nd row: reconstruction of normal data; 3rd row: anomalous data; 4th row: reconstruction of anomalous data.

Images are from Zenati et al. (2018).

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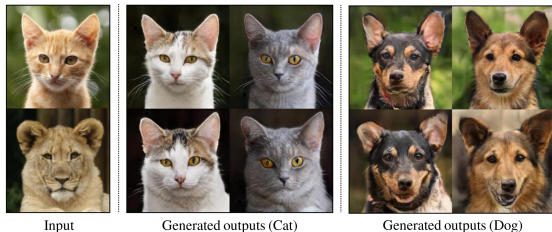
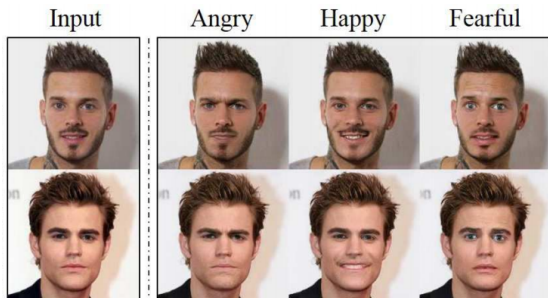
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Data-to-Data Translation

Conditioning Data (C)	Target Data (X)	Name of Corresponding Task
Image	Image	Image-to-Image translation
Video	Video	Video-to-Video translation
Text	Text	Text-to-Text translation
Text	Image	Text-to-Image translation

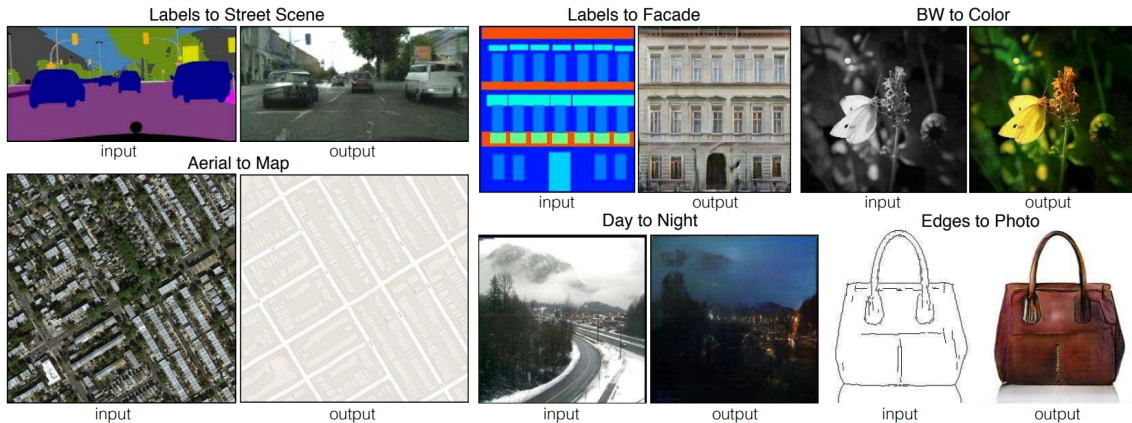
- Data-to-Data translation refers to synthesizing one type of data (from the *source domain*) using another (from the *target domain*).

Data-to-Data Translation



Images are edited from Choi et al. (2018) (left) and Choi et al. (2020) (right).

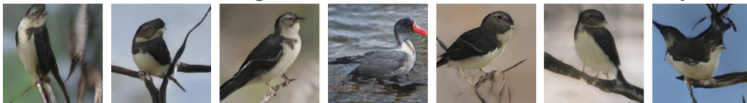
Data-to-Data Translation



Summary of applications of Image-to-Image translation (Isola et al., 2017).

Data-to-Data Translation

this bird has wings that are **black** and has a **white** belly



this bird has wings that are **red** and has a **yellow** belly

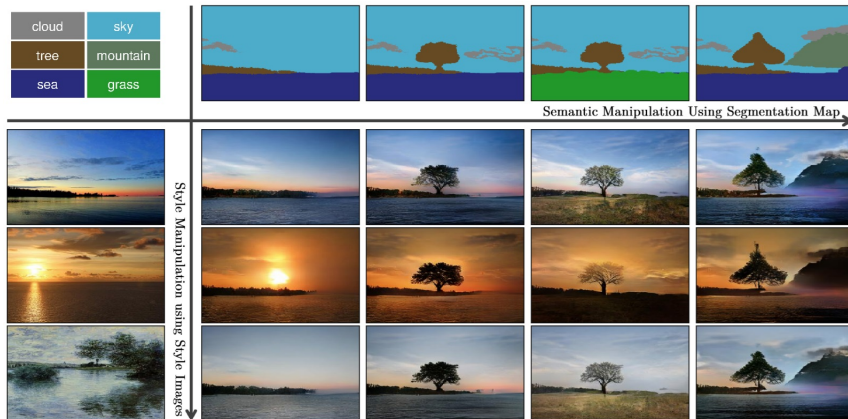


this bird has wings that are **blue** and has a **red** belly



Images are from Xu et al. (2018).

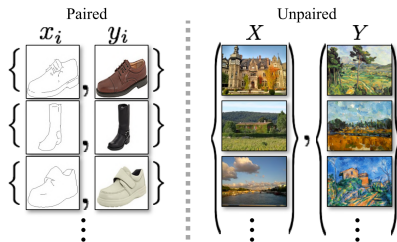
Data-to-Data Translation



- Data-to-Data Translation is an example of *Controllable Generation*. Changes in the conditioning data influence the specific characteristics of the generated results.

Images are from Park et al. (2019).

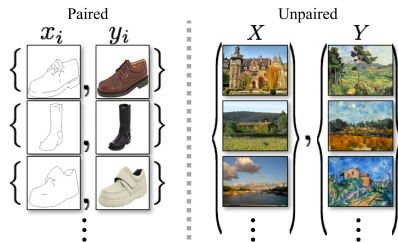
Paired vs. Unpaired Datasets



- ▶ According to the presence of pair information, methods are divided into unpaired and paired settings.
- ▶ A dataset is *paired* if both source and target data (modalities) are observed for every subject, and pair information (ID of subjects) is provided. If the pair information is not provided, the dataset is *unpaired*.

The figure is from Zhu et al. (2017).

Paired vs. Unpaired Datasets



- ▶ The pair of a given source datum usually indicates the most similar datum in the target dataset. The criterion of 'similar' depends on the context.
- ▶ Especially for unpaired datasets, it is difficult to define pairs. For example, when the source and target data are horse and zebra images, respectively, no object possesses the phenotype of both a horse and a zebra simultaneously.

The figure is from Zhu et al. (2017).

Paired vs. Unpaired Datasets

- ▶ Earliest works usually assume a one-to-one relationship between source and target data, regardless of whether the training dataset contains the pair information.
- ▶ The translator is referred to as a “generator” and denoted by $G_{\theta}(\mathbf{C})$ since it learns the data distribution ($p(\mathbf{x})$), but its role is more of a transformer than a generator.
- ▶ However, the mapping from source data to target data is often one-to-many. For example, when we translate daytime scenes to their nighttime counterparts, there can be multiple nighttime scenes for a given daytime scene.
- ▶ Recent works have introduced latent factors $\mathbf{Z}$ to extend the translator $G_{\theta}(\mathbf{C})$ to $G_{\theta}(\mathbf{Z}, \mathbf{C})$. This research field is often called *multimodal synthesis*.

Paired Data-to-Data Translation

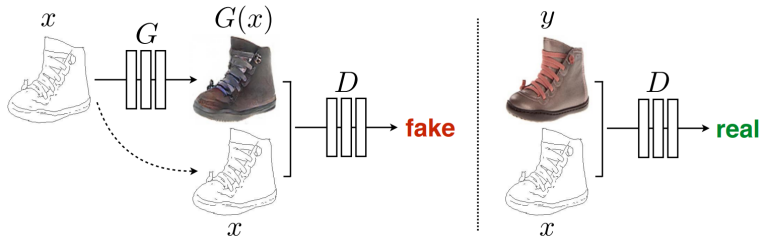


Figure: The architecture of Pix2Pix (Isola et al., 2017).

- ▶ We denote paired datasets by $\{(\mathbf{x}_i, \mathbf{c}_i)\}_{i=1}^n$. The generator G_θ inputs $\mathbf{c}_i$ to synthesize datum similar to $\mathbf{x}_i$.
- ▶ This can be viewed as an multivariate regression utilizing adversarial learning.

Paired Data-to-Data Translation

- ▶ The goal is to learn a translator G_θ s.t. $G_\theta(\mathbf{C}) = \mathbf{X}$. For example, in Pix2Pix (Isola et al., 2017), the value function is:²

$$V(\theta, \phi) := \int (\log D_\phi(\mathbf{c}, \mathbf{x})) p(\mathbf{c}, \mathbf{x}) d\mathbf{c} d\mathbf{x} + \int (\log (1 - D_\phi(\mathbf{c}, G_\theta(\mathbf{c})))) p(\mathbf{c}) d\mathbf{c}. \quad (15)$$

- ▶ The objective function is:

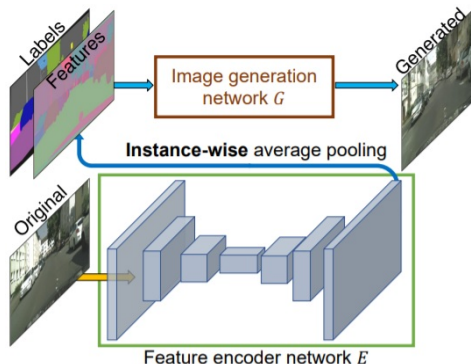
$$\int \|\mathbf{x} - G_\theta(\mathbf{c})\|_1 p(\mathbf{c}, \mathbf{x}) d\mathbf{c} d\mathbf{x} + \lambda V(\theta, \phi) \quad (16)$$

where the first term is the prediction error and λ is a hyperparameter.

- ▶ Since the objective includes $p(\mathbf{c}, \mathbf{x})$, we need paired data to use it.

²The authors applied dropout, a stochastic layer, to build a probabilistic G_θ . For simplicity, we omit this stochastic part in the formulation in this lecture.

Paired Data-to-Data Translation



- ▶ Pix2PixHD (Wang et al., 2018) is an extension of Pix2Pix and introduces a feature encoder network to address multi-modal synthesis problem.

The figure is from Wang et al. (2018).

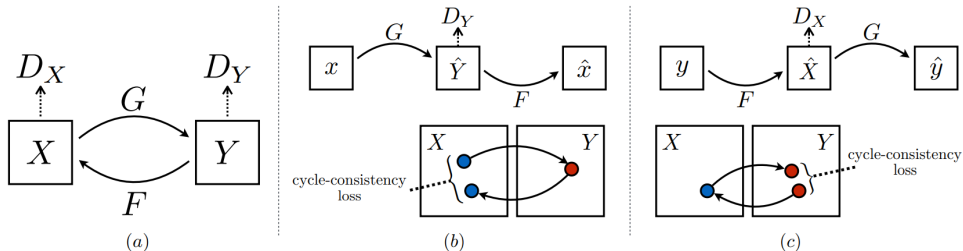
Paired Data-to-Data Translation

- ▶ The value function is:

$$\begin{aligned} V(\theta, \phi, \psi) := & \int (\log D_{\phi}(\mathbf{c}, \mathbf{x})) p(\mathbf{c}, \mathbf{x}) d\mathbf{c} d\mathbf{x} \\ & + \int \left(\log (1 - D_{\phi}(\mathbf{c}, G_{\theta}(\mathbf{c}, E_{\psi}(\mathbf{x}))) \right) p(\mathbf{c}, \mathbf{x}) d\mathbf{c} d\mathbf{x}. \end{aligned} \quad (17)$$

- ▶ Pix2PixHD introduces latent variables, but their distribution is not estimated.
- ▶ The K -means clustering algorithm is applied to latent variables extracted from the training dataset. At inference time, one of K centroids is randomly sampled to generate target data.

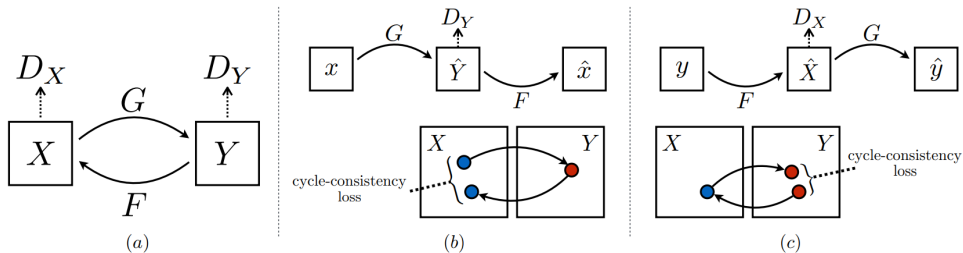
Unpaired Data-to-Data Translation



- ▶ The pair information is usually not provided.
- ▶ CycleGAN (Zhu et al., 2017) is one of the earliest unpaired data-to-data translation models. It trains two translators and utilizes the cycle-consistency loss.

The figure is from Zhu et al. (2017).

Unpaired Data-to-Data Translation



- ▶ CycleGANs have a pair of translators: one from the source domain to the target ($G_{C \rightarrow X, \theta}(\mathbf{c})$) and the other from the target back to the source ($G_{X \rightarrow C, \theta}(\mathbf{x})$).
- ▶ The goal is to find ($G_{C \rightarrow X, \theta}, G_{X \rightarrow C, \theta}$):
 1. (Transportation) $p_{\theta}(G_{C \rightarrow X, \theta}(\mathbf{c})) = p(\mathbf{x})$ and $p_{\theta}(G_{X \rightarrow C, \theta}(\mathbf{x})) = p(\mathbf{c})$.
 2. (Cycle-Consistency) $(G_{C \rightarrow X, \theta} \circ G_{X \rightarrow C, \theta})(\mathbf{c}) = \mathbf{c}$ and $(G_{X \rightarrow C, \theta} \circ G_{C \rightarrow X, \theta})(\mathbf{x}) = \mathbf{x}$.

The figure is from Zhu et al. (2017).

Unpaired Data-to-Data Translation

- ▶ The value function can be expressed as:

$$V(\theta, \phi) := \left(\int (\log D_{X,\phi}(\mathbf{x})) p(\mathbf{x}) d\mathbf{x} + \int (\log (1 - D_{X,\phi}(G_{C \rightarrow X, \theta}(\mathbf{c}))) p(\mathbf{c}) d\mathbf{c} \right) \\ + \left(\int (\log D_{C,\phi}(\mathbf{c})) p(\mathbf{c}) d\mathbf{c} + \int (\log (1 - D_{C,\phi}(G_{X \rightarrow C, \theta}(\mathbf{x}))) p(\mathbf{x}) d\mathbf{x} \right).$$

- ▶ They introduce cycle-consistency loss:

$$\int \| (G_{C \rightarrow X, \theta} \circ G_{X \rightarrow C, \theta})(\mathbf{c}) - \mathbf{c} \|_1 p(\mathbf{c}) d\mathbf{c} + \int \| (G_{X \rightarrow C, \theta} \circ G_{C \rightarrow X, \theta})(\mathbf{x}) - \mathbf{x} \|_1 p(\mathbf{x}) d\mathbf{x}.$$

The cycle-consistency loss encourages the inverse relation between two translators.

- ▶ The loss function also includes the identity mapping loss term:

$$\int \| \mathbf{x} - G_{X \rightarrow C, \theta}(\mathbf{x}) \| p(\mathbf{x}) d\mathbf{x} + \int \| \mathbf{c} - G_{C \rightarrow X, \theta}(\mathbf{c}) \| p(\mathbf{c}) d\mathbf{c}.$$

This approach can be viewed as learning pair information by neural networks.

Unpaired Data-to-Data Translation

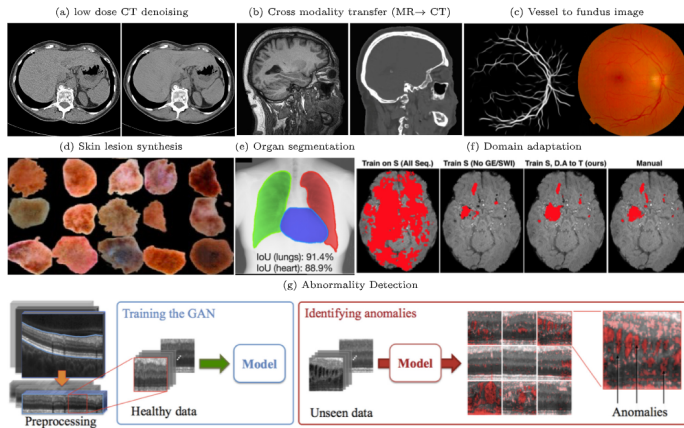


Figure: Important applications include image reconstruction, segmentation, detection, classification, and cross-modality synthesis (Yi et al., 2019)

Unpaired Data-to-Data Translation: Cross modality transfer (MR $\rightarrow$ CT)

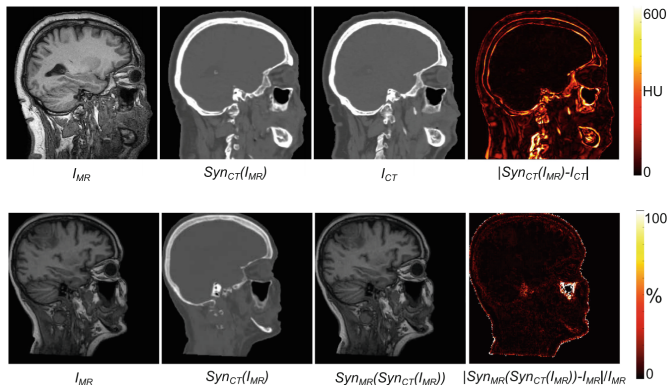


Figure: Wolterink et al. (2017) trained CycleGAN to translate MR images into CT images. The translation from CT to MR ((2, 2) $\rightarrow$ (2, 3)) is also successful.

Table of Contents

1. Generative Adversarial Network
2. Representation Learning with Adversarial Learning
3. Data-to-Data Translation
4. Extension to f -divergence

f -GAN

- ▶ We have reviewed the following relationship in GANs, which holds up to a constant addition and sign-preserving multiplication: Using optimal discriminators, we have:

$$\mathcal{D}_{\text{JS}}(p \parallel p_\theta) \approx V(\theta, \hat{\phi}_n(\theta)).$$

- ▶ Nowozin et al. (2016) generalized the concept of using auxiliary networks to approximate other f -divergences:

$$\mathcal{D}_f(p \parallel q) := \int f\left(\frac{p(\mathbf{x})}{q(\mathbf{x})}\right) q(\mathbf{x}) d\mathbf{x} \quad (18)$$

where $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is a convex function satisfying $f(1) = 0$.

- ▶ The key idea is to introduce the convex conjugate function (or Fenchel conjugate, Hiriart-Urruty and Lemaréchal, 2004) to derive variational estimations of f -divergences.

f -Divergence

- ▶ Examples include KL divergence, total variation distance, and Jensen-Shannon (JS) divergence.

- ▶ **KL divergence:** $\mathcal{D}_{\text{KL}}(p||q) := \int \log(p(\mathbf{x})/q(\mathbf{x}))p(\mathbf{x})d\mathbf{x}$

Proof: When $f(u) = u \log u$,

$$\mathcal{D}_f(p||q) = \int \left(\frac{p(\mathbf{x})}{q(\mathbf{x})} \log \frac{p(\mathbf{x})}{q(\mathbf{x})} \right) q(\mathbf{x}) d\mathbf{x} = \int \left(\log \frac{p(\mathbf{x})}{q(\mathbf{x})} \right) p(\mathbf{x}) d\mathbf{x}.$$

- ▶ **Total variation distance:** $\delta(p, q) := \frac{1}{2} \int |p(\mathbf{x}) - q(\mathbf{x})| d\mathbf{x}$

Proof: When $f(u) = |u - 1|/2$,

$$\mathcal{D}_f(p||q) = \frac{1}{2} \int \left| \frac{p(\mathbf{x})}{q(\mathbf{x})} - 1 \right| q(\mathbf{x}) d\mathbf{x} = \frac{1}{2} \int |p(\mathbf{x}) - q(\mathbf{x})| d\mathbf{x}.$$

f -Divergence

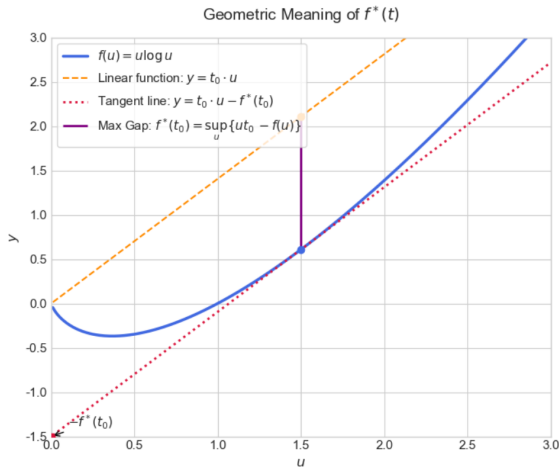
- ▶ Jensen-Shannon divergence:

$$\mathcal{D}_{\text{JS}}(p||q) := \frac{1}{2}(\mathcal{D}_{\text{KL}}(p||\frac{p+q}{2}) + \mathcal{D}_{\text{KL}}(q||\frac{p+q}{2})) \quad (19)$$

f -Divergence

Name	$D_f(P\ Q)$	$f(u)$
Kullback-Leibler	$\int p(x) \log \frac{p(x)}{q(x)} \mathrm{d}x$	$u \log u$
Reverse KL	$\int q(x) \log \frac{q(x)}{p(x)} \mathrm{d}x$	$-\log u$
Pearson χ^2	$\int \frac{(q(x)-p(x))^2}{p(x)} \mathrm{d}x$	$(u-1)^2$
Squared Hellinger	$\int \left(\sqrt{p(x)} - \sqrt{q(x)} \right)^2 \mathrm{d}x$	$(\sqrt{u} - 1)^2$
Jensen-Shannon	$\frac{1}{2} \int p(x) \log \frac{2p(x)}{p(x)+q(x)} + q(x) \log \frac{2q(x)}{p(x)+q(x)} \mathrm{d}x$	$-(u+1) \log \frac{1+u}{2} + u \log u$

List of popular examples of f -divergences, edited from Nowozin et al. (2016).

f -GAN

- We denote the convex conjugates of functions f by $f^*(t) := \sup_u \{ut - f(u)\}$.

f -GAN

- ▶ The f^* relates f and its subgradients. When f is convex and differentiable³, the following properties hold:
 1. f^* is also convex and differentiable.
 2. Duality holds, i.e., $(f^*)^* = f$.
 3. The relation $f(u) + f^*(t) = ut$ holds if and only if $t = f'(u)$.
- ▶ When f' is invertible, $f^*(t) = (f')^{-1}(t)t - (f \circ f'^{-1})(t)$.

³For more general functions, check Hiriart-Urruty and Lemaréchal (2004).

f -GAN

- By duality and the definition of supremum, we have the following variational formulation:

$$\begin{aligned}\mathcal{D}_f(p||p_\theta) &= \int \sup_t \left\{ \frac{p(\mathbf{x})}{p_\theta(\mathbf{x})} t - f^*(t) \right\} p_\theta(\mathbf{x}) d\mathbf{x} \\ &\geq \sup_{T_\phi} \left(\int T_\phi(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} - \int f^*(T_\phi(\mathbf{x})) p_\theta(\mathbf{x}) d\mathbf{x} \right).\end{aligned}\tag{20}$$

- We assume that f is differentiable and that $\{T_\phi | \phi \in \Phi\}$ is sufficiently flexible such that for any given θ , there exists $\phi^*(\theta)$ where:

$$T_{\phi^*(\theta)}(\mathbf{x}) = f' \left(\frac{p(\mathbf{x})}{p_\theta(\mathbf{x})} \right).\tag{21}$$

This satisfies the equality condition of Equation (20).

f -GAN

- ▶ Define $F(\theta, \phi) := \int T_\phi(\mathbf{x})p(\mathbf{x}) d\mathbf{x} - \int f^*(T_\phi(\mathbf{x}))p_\theta(\mathbf{x}) d\mathbf{x}$ and $\theta^* := \arg \min_{\theta} (\max_{\phi} F(\theta, \phi))$.
- ▶ Then, $\max_{\phi} F(\theta, \phi) = F(\theta, \phi^*(\theta)) = \mathcal{D}_f(p \| p_\theta)$. Thus, p_{θ^*} is a minimizer of the f -divergence.
- ▶ **Example 1 (KL divergence):** Let $f(u) = u \log u$ and $f^*(t) = \exp(t - 1)$. We can express $F(\theta, \phi)$ as follows:

$$F(\theta, \phi) = \int T_\phi(\mathbf{x})p(\mathbf{x}) d\mathbf{x} - \int \exp(T_\phi(\mathbf{x}) - 1)p_\theta(\mathbf{x}) d\mathbf{x}. \quad (22)$$

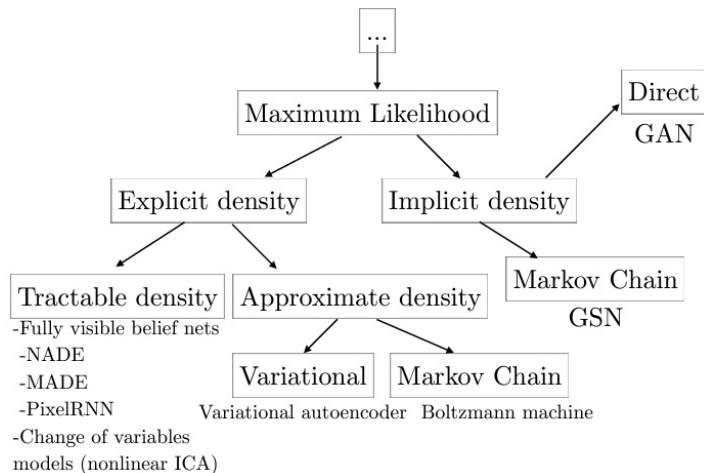
f -GAN

- **Example 2 (JS divergence):** Let $f(u) = -(u+1) \log \frac{1+u}{2} + u \log u$ and $f^*(t) = -\log(2 - \exp(t))$. We can express $F(\theta, \phi)$ as follows:

$$F(\theta, \phi) = \int T_{\phi}(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} + \int \log \left(2 - \exp \left(T_{\phi}(\mathbf{x}) \right) \right) p_{\theta}(\mathbf{x}) d\mathbf{x}. \quad (23)$$

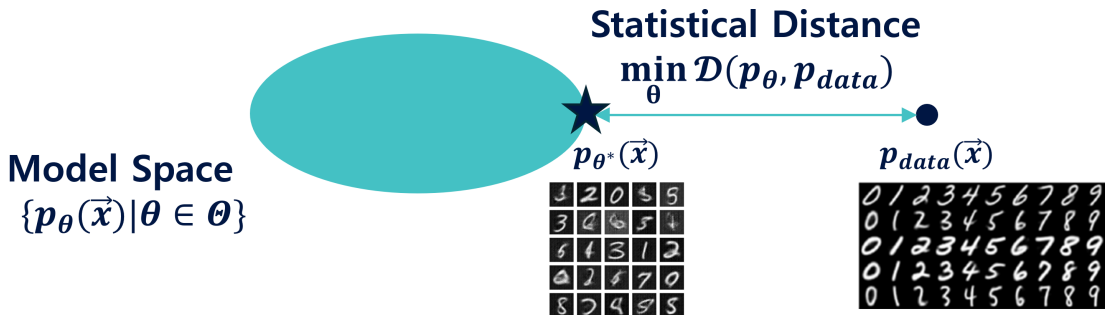
- GANs are special cases of f -GANs. When we model $T_{\phi}(\mathbf{x}) = \log D_{\phi}(\mathbf{x}) + \log 2$, $F(\theta, \phi) = V(\theta, \phi) + \log 4$, and $T_{\phi^*(\theta)}(\mathbf{x}) = \log D_{\phi^*(\theta)}(\mathbf{x}) + \log 2 = \log \frac{p(\mathbf{x})}{p(\mathbf{x}) + p_{\theta}(\mathbf{x})} + \log 2$ hold.

Summary



The figure is from Goodfellow (2016).

Summary



Handwritten digit images are from Wolterink et al. (2017).

Summary

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
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8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

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